

Westravelle Analytical Ltd

APPROVED 11 MAY 2012

STABILITY REPORT: EP19 CAPSULE BLEND, LOT L02

Six-month accelerated data

Protocol: WA/2011/192

Client: Esmervane Pharmaceuticals

Report date: 07/05/2012

1. SCOPE

1.1 This report gives results for lot L02 of the EP19 capsule blend stored under protocol WA/2011/192 for up to six months. Storage started on 07/11/2011. 1.2 Conclusions are limited to the tested interval and conditions.

2. MATERIALS AND PACKAGING

- 2.1 Stability lot: EP19 capsule blend, lot L02, filled into size 1 hard gelatin capsules
- 2.2 Pack: 60 cc HDPE bottle, induction-sealed, 1 g silica gel desiccant, 30 capsules per bottle
- 2.3 Storage: 25 °C/60% RH, 30 °C/65% RH and 40 °C/75% RH in mapped chambers

3. METHODS

- 3.1 Assay and related substances: HPLC method WA-AM-0412 Rev 02 (qualified 10/2011), UV 254 nm
- 3.2 Water: Karl Fischer coulometric, WA-AM-0105
- 3.3 System suitability per sequence: resolution Imp A / main peak ≥ 2.0 , tailing ≤ 1.5 , %RSD of six standard injections $\leq 1.0\%$

3.4 All sequences in this report met system suitability.

4. SPECIFICATION

4.1 Assay 95.0-105.0% of label claim; Imp A (RRT 0.86, specified) $\leq 0.50\%$; Imp B (RRT 1.12, specified) $\leq 0.30\%$; any unspecified impurity $\leq 0.20\%$; total impurities $\leq 2.0\%$.

5. RESULTS

5.1 Initial assay 99.4% of label claim. Results to 6 months are tabulated below.

Table 1 Stability results

Condition	Timepoint (months)	Assay (% LC)	Imp A RRT 0.86 (%)	Imp B RRT 1.12 (%)	Total impurities (%)	Water KF (% w/w)	Result
25 °C/60% RH	0	99.5	0.04	<0.05	0.17	2.1	Complies
25 °C/60% RH	1	99.3	0.06	<0.05	0.22	1.9	Complies
25 °C/60% RH	3	99.3	0.08	<0.05	0.24	1.1	Complies
25 °C/60% RH	6	99.1	0.10	<0.05	0.26	1.4	Complies
30 °C/65% RH	0	99.4	0.05	<0.05	0.21	2.1	Complies
30 °C/65% RH	1	99.3	0.06	<0.05	0.15	1.0	Complies
30 °C/65% RH	3	99.0	0.10	0.07	0.31	2.2	Complies
30 °C/65% RH	6	99.2	0.14	0.09	0.30	1.2	Complies
40 °C/75% RH	0	99.2	0.05	<0.05	0.22	2.4	Complies
40 °C/75% RH	1	99.7	0.09	0.06	0.26	2.2	Complies
40 °C/75% RH	3	98.9	0.21	0.12	0.38	1.7	Complies
40 °C/75% RH	6	97.7	0.35	0.21	0.69	1.4	Complies

Specification: assay 95.0-105.0% LC; Imp A (RRT 0.86, specified) $\leq 0.50\%$; Imp B (RRT 1.12, specified) $\leq 0.30\%$; any unspecified impurity $\leq 0.20\%$; total impurities $\leq 2.0\%$

6. DISCUSSION

6.1 All results comply with the specification. Assay and impurities are essentially unchanged at 25 °C/60% RH and 30 °C/65% RH. 6.2 At 40 °C/75% RH, Imp A increases with time. Water content does not show a matching upward trend, so moisture uptake through the pack is not demonstrated; hydrolysis of the amide is a hypothesis that should be confirmed by identifying the degradant and by a humidity-controlled comparison. 6.3 A tighter moisture barrier should be evaluated before long-term testing.

7. DEVIATIONS

7.1 None.

8. RECORDS

8.1 Sequence sheets, chromatograms and chamber records are held under WST-11-005959.

Name	Role	Signature	Date
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